Paleomagnetism of Metamorphic Rocks of Daly Bay and Melville Peninsula, Churchill Structural Province[1]

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Studies of metamorphic rocks from the Daly Bay and Melville Peninsula areas of the Churchill Structural Province of the Canadian Shield have yielded two groups of directions. Group *A* directions (16 sites) from gneisses of basic and intermediate composition have the mean 196°, +55° ($N = 16$, $k = 39$, $\alpha_{95} = 6.1°$) with a corresponding pole at 12°N, 101°W ($k = 27$, $\alpha_{95} = 7°$). These rocks are foliated and the average effect of anisotropy on the high coercive force (stable) components is small. Evidence is presented to show that agewise the pole falls within the range of 1250 to 2100 m.y. The pole lies on the apparent polar wander curve for the Canadian Shield in the time range 1425 to 1725 m.y. which defines its age within narrower limits. Since the mean K–Ar age is 1622 ± 30 m.y., the magnetization of Group *A* may reflect this age. This result shows that metamorphic rocks can be used to provide useful paleomagnetic data. The Group *B* magnetization, however, derived mainly from acidic gneisses and having a mean direction of 055°, +72° ($N = 8$, $k = 18$, $\alpha_{95} = 13°$) and a pole at 67°N, 009°W very close to the present pole, is probably of recent origin.

L'étude des roches métamorphiques de la Province Structurale de Churchill (Bouclier Canadien) dans la région de la Baie de Daly et de la Péninsule de Melville a permis de délimiter deux groupes de directions. Les directions du groupe *A* furent obtenues à partir de gneiss de composition mafique et intermédiaire et ont une moyenne de 196°, +55° ($N = 16$, $k = 39$, $\alpha_{95} = 6.1°$) avec un pôle correspondant de 12°N, 101°W ($k = 27$, $\alpha_{95} = 7°$). Ces roches sont foliées et l'effet de cette anisotropie sur les composantes de la force compressive (stable) fut minime. Les évidences permettent de dire qu'au point de vue temps, les pôles tombent entre les dates de 1250 m.a. et 2100 m.a. Les pôles sont situés sur la courbe apparente de dérive du pôle du Bouclier Canadien pour l'intervalle allant de 1425 m.a. à 1725 m.a. Ceci permet donc de préciser encore plus leur âge. Vu que l'âge moyen par K–Ar est de 1622 ± 30 m.a., la magnétisation du groupe *A* peut correspondre à cet âge. Les résultats obtenus démontrent que les roches métamorphiques peuvent fournir d'importantes données paléomagnétiques. La magnétisation du groupe *B*, qui fut obtenue à partir de gneiss acides, à une direction moyenne de 055°, +72° ($N = 8$, $k = 18$, $\alpha_{95} = 13°$) et un pôle à 67°N, 009°W, pôle qui est très près du pôle actuel et qui pourrait bien etre d'origine récente.

[Traduit par le journal]

Introduction

Hitherto, most paleomagnetic data have been obtained from igneous and sedimentary rocks. Workers have tended to shy away from metamorphic rocks because of possible errors in directions due to the presence of linear or planar fabric that might deflect the remanence away from the ambient field. This paper gives an account of a study of magnetization directions in gneissic rocks that was made in order to determine whether or not the presence of an anisotropic fabric constitutes a severe impediment to their use as indicators of the paleomagnetic field.

Geology and Sampling

Samples were collected during 1971 from the southern Melville Peninsula (Repulse Bay area) and from the Daly Bay area (Fig. 1(A)). The rocks sampled have been folded, metamorphosed, and intruded by granitic and pegmatitic rocks. Several units (Heywood 1967) have been recognized; for example, the Daly Bay Complex, which is an assemblage of anorthosites, gabbros, gneisses, and migmatites. Twelve K–Ar age determinations (11 biotite, 1 muscovite) on well-distributed samples from four map units (9a, 10, 12, 15 of Fig. 1. B and C) yield ages ranging from 1580 to 1690 m.y. with a mean of 1622 ± 30 m.y. (Heywood 1967, Wanless *et al.* 1967). As interpreted by Heywood (in Wanless *et al.* 1967), these ages

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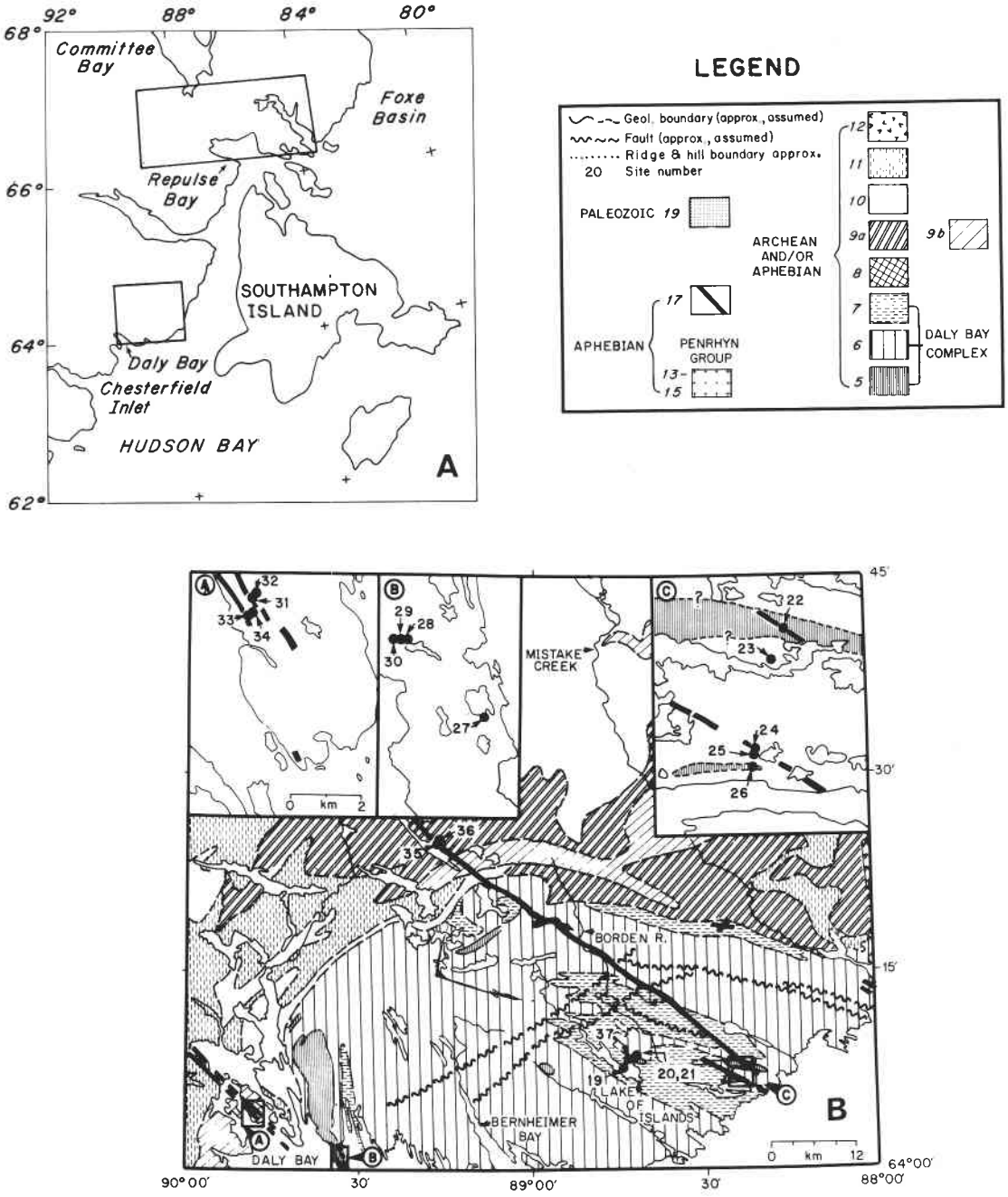
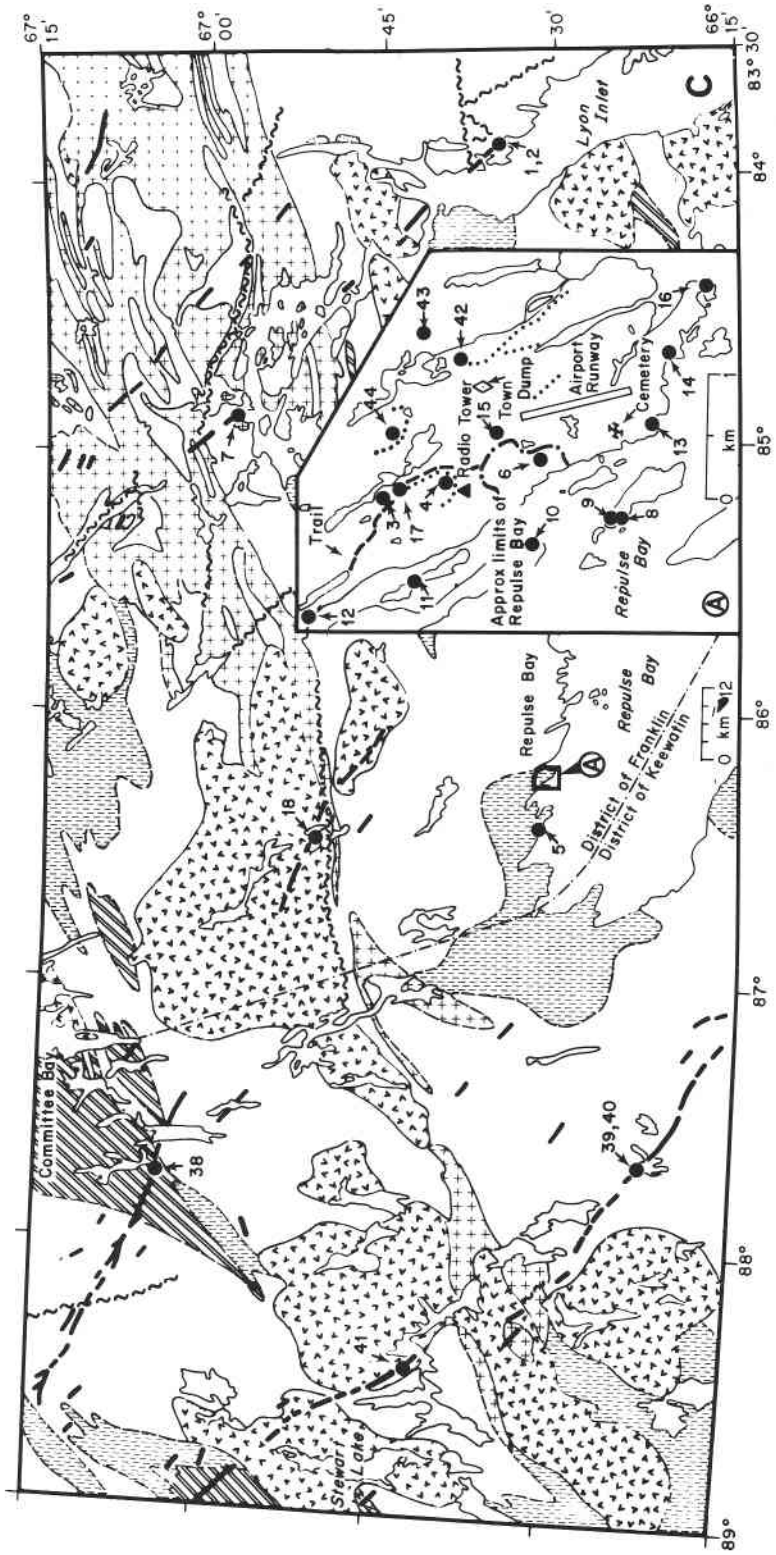


FIG. 1. A. Map showing location of Daly Bay and Repulse Bay areas. B. Geology and sampling sites of the Daly Bay area. C. (See facing page.) Geology and sampling sites of the Repulse Bay area.

The map units numbered in the legend (after Heywood 1967) are as follows: 5. anorthosite and gabbro, 6. gneiss and layered gneiss, minor amphibolite, granulite, and charnockite, 7. migmatite, 8a. peridotite and serpentinite, 9a. layered gneiss, 9b. paragneiss and schist, 10. granitic gneiss, 11. migmatite, 12. granite granodiorite, and quartz monzonite, 13–15. paragneiss, schist, crystalline limestone, and quartzite, 17. diabase dikes and sills, 19. dolomite and minor sandstone.


67° 15'
67° 00'
45'
30'
66° 15'
83° 30'
84°
85°
86°
87°
88°
89°
Lyon Inlet
C
1,2
Repulse Bay
Repulse Bay
District of Franklin
District of Keewatin
0 km 12
Committee Bay
Stewart Lake
38
39,40
41
18
5
7
Trail
Approx limits of Repulse Bay
Radio Tower
Town Dump
Airport Runway
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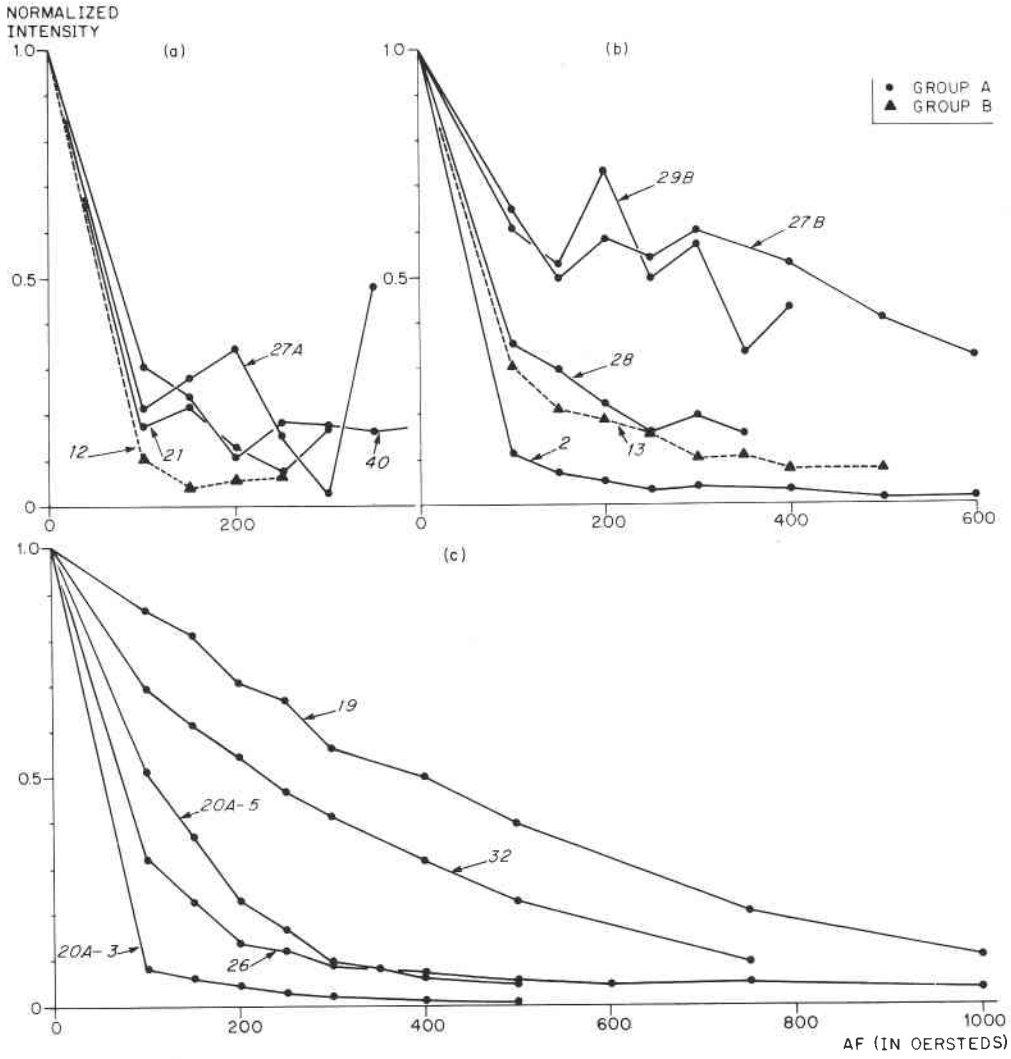


FIG. 2. Examples of a.f. demagnetization curves of unacceptable (a) and acceptable (b and c) specimens. Those specimens which have temporary components in the cleaning field range and those which have definite end-points are shown in (b) and (c) respectively.

correspond either to the last metamorphism or to the granite (map unit 12) emplacement. The basement is cut by large northwest-trending unmetamorphosed posttectonic diabase dikes, the paleomagnetism of which will be reported later.

Natural Remanent Magnetization

Three to five samples (cores or hand samples) were obtained from each site. Sample averages (mean of 2 specimens from each sample) are the basic units in the statistical analysis. Specimens were cleaned in alternating fields (a.f.) in the apparatus described by Roy *et al.* (1969, 1973 (in preparation)) and measured on a commercial spinner magnetometer. Cleaning fields were determined using standard procedures (As and Zijderveld 1958, Irving *et al.* 1961).

Samples not fulfilling certain criteria are not used in the analysis. These criteria are as follows: (a) the difference between specimens from the same sample $\geqslant 45°$ ($R \leqslant 1.848$), (b) magnetometer readings changing during mea-

TABLE 1. Site statistics

Site	Rock type	UTMG	*N*	a.f. (Oe)	$D°, I°$ (*R*)
Group *A*					
1	metadiabase	17WLD7389	5	500	219, +50 (4.64)
2	metadiabase	17WLD7389	5	200	211, +54 (4.76)
18	bi–hb–g–gn	16WEK2316	3 (2)	100	216, +55 (2.95)
19	gar–bi–st–gn	16WDG1412	5	750	193, +53 (4.94)
20A	metadiabase	16WDG1714	5	250	177, +58 (4.99)
20B	meta-pyx–gran	16WDG1714	3 (2)	300	219, +72 (2.84)
37	metagabbro	16WDG1513	3	250	174, +51 (2.93)
23	hb–pla–gran	16WDG3112	3	250	202, +44 (2.99)
25	amphibolite	16WDG3110	3	400	194, +34 (2.96)
26	anorthosite	16WDG3110	7	250	176, +41 (6.95)
27B	bi–pla–hb–gn	16WCG7502	3 (2)	150	200, +50 (2.86)
28	hb-bearing sheared an	16WCG7404	3 (2)	150	185, +63 (2.93)
29B	anorthosite	16WCG7404	4	200	176, +66 (3.96)
30	anorthosite	16WCG7404	5	250	211, +55 (4.96)
32	bi–hb–g–gn	16WCG6208	5	300	198, +57 (5.00)
34	bi–g–gn	16WCG6208	5	300	198, +61 (4.96)
Group *B*					
10	g/g–gn	16WEJ3378	2 (1)	250	015, +77 (2.00)
11	g/g–gn	16WEJ3379	2 (1)	150	096, +86 (1.97)
13	g/g–gn	16WEJ3477	2 (1)	200	034, +51 (1.96)
14	g/g–gn	16WEJ3577	2 (1)	150	069, +39 (1.96)
42	g/g–gn	16WEJ3579	3	200	091, +79 (2.90)
7D	metadiabase	16WEK9231	9 (1)	200	093, +84 (8.71)
7C	amphibolite/g–gn	16WEK9231	6 (7)	200	078, +71 (5.60)
41C	hb–bi–g–gn	16WDJ3699	6	200	011, +75 (5.86)

NOTES: The abbreviations under 'Rock type' are: (an) anorthosite, (bi) biotite, (g) granitic or granite, (gar) garnet, (gn) gneiss, (gran) granulite, (hb) hornblende, (pla) plagioclase, (pyx) pyroxene, and (st) sillimanite. UTMG is the Universal Transverse Mercator Grid reference. The number of acceptable and unacceptable (in brackets) samples (cores or handsamples) are given under *N*. a.f. is the cleaning field in oersteds. $D°$, $I°$ are the declination and inclination (with respect to horizontal and true north) of the mean site directions (or mean Group directions in Table 2) after cleaning. *R* (in brackets) is the vector resultant.

surement, and (c) directions of samples from the same site were random (criteria of Vincenz and Bruckshaw 1960) both before and after cleaning. Unacceptable samples, which are mostly granitic gneisses, constitute 52% of the collection.

The remaining samples are more basic in composition (Table 1). Most of them exhibit clearly defined end-points (decay curves Fig. 2(c)) and are magnetically stable, but a number show appreciable scatter in the cleaning field range (Fig. 2(b)). This scatter is random and due mainly to temporary components acquired in the laboratory.

After magnetic cleaning the accepted samples show two groups of directions referred to as *A* and *B*. Group *A* directions (in 67 samples from 16 sites) are directed towards the south, and Group *B* (in 32 samples from 8 sites) towards the northeast (Table 2 and Fig. 3). Group *A* directions are widely distributed areally and stratigraphically in map units 5, 6, 9, 10, 11, and 12 (essentially basic to intermediate composition gneisses) of Heywood (1967) (Fig. 1(B), 1(C)); whereas Group *B* directions are confined to map units 10 to 12 (essentially granitic gneisses) in the Repulse Bay area (Fig. 1(B)). Group *A* magnetizations are in general more stable than those of Group *B*, temporary components being less in evidence and coercivities (or median destructive fields) of n.r.m. (natural remanent magnetization) being more than twice as large. The presence of higher coercivities of i.r.m. (isothermal remanent magnetization) in the *B* specimens (Table 3) than in the *A* specimens is a puzzle. There could be many explanations. For example, *B* magnetizations might be composed of two components whose directions differ by a large angle. If so, the Group *B* directions are not reliable indicators of the Precambrian geomagnetic field. Because of this uncertainty, and because the mean direction does not differ significantly from the present

TABLE 2. Summary of results

Entry	Description	N	$D°, I°$	k	α_{95}	Pole
1	Group *A* directions	16	196, +55	38	6.0 (6.1)	11N,101W
2	Group *A* poles	16	—	27	7.2	12N,101W
3	Group *B* directions	8	055, +72	18	13.3 (13.4)	64N,003W
4	Group *B* poles	8	—	10	18.7	67N,009W

NOTES: N is the number of sites, and k is Fisher's index of precision (Fisher 1953). The radii of the 95% circles of confidence (α_{95}) both uncorrected and corrected (in brackets) for anisotropy effects are given. The semi-axes (δ_p, δ_m) of the ellipses of confidence ($p = 0.05$) for entries 1 and 3 are (6°, 8°) and (21°, 24°) respectively. The latitude and longitude of the Group poles are given in the last column. Other symbols are as in Table 1.

TABLE 3. Summary of some magnetic properties

	Intensities		Coercivities	
Description	n.r.m.	i.r.m.	n.r.m.	i.r.m.
Group *A*	164 ± 9	5.9 ± 28.5	113 ± 2	188 ± 2
Group *B*	87.6 ± 2.9	8.9 ± 2.5	43 ± 2	237 ± 2

NOTES: The log mean intensities of n.r.m. and i.r.m. are in c.g.s. $cm^{-3} \times 10^{-6}$ and c.g.s. $cm^{-3} \times 10^{-3}$ respectively. Coercivities (median destructive fields) are given in oersteds. Averages of one specimen per site.

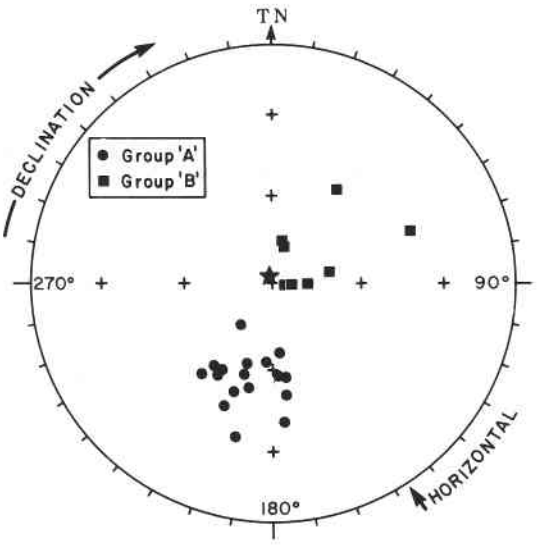


FIG. 3. Site means plotted. All positive inclinations. A star denotes the present earth's field.

field, the Group *B* results are not considered further.

One site (20A, a metamorphosed dike segment) has two samples (4 specimens) in an inexact reversed configuration and three (6 specimens) grouped close to the mean cleaned direction (Fig. 4). Cleaning in 100 Oe removes nearly all of the reversed component. The average directions of the n.r.m. of the reversed specimens all lie near a great circle passing through the mean cleaned normal direction and its opposite. This is the only evidence for reversals of magnetization within Group *A*.

The significance and stability of the Group *A* direction are demonstrated by the variety of rocks with this direction (Table 1), the good grouping of the site directions (Fig. 2) and their departure from the recent earth's field direction, and the evidence for reversal.

An upper limit for the age of the Group *A* magnetization is set by studies of the post-tectonic diabase dikes (Park 1973, in preparation). These dikes belong to the Mackenzie swarm (Fahrig and Jones 1969). Studies of basement rocks near the dikes show that adjacent to the dikes the magnetization of the baked rocks is approximately parallel to the dike but at a distance (>10–50 m, depending on the dike width) the Group *A* direction of the country rocks remains unaltered. Thus the Group *A* magnetization predates the Mackenzie diabase which Fahrig and Jones (1969) estimated to be 1250 m.y. old. A lower limit for the age of magnetization is more difficult to establish. It would presumably coincide with the age of regional metamorphism (amphibolite facies) during which recrystallization and heating (about 600 °C) would be expected to have destroyed any earlier magnetization. Such an age would be reflected by Rb–Sr isochrons.

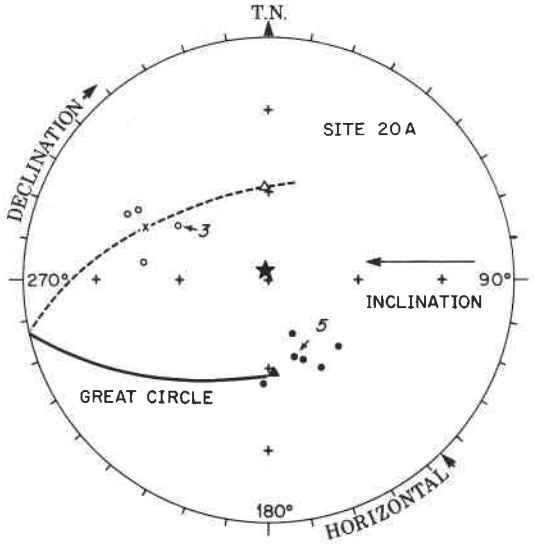


FIG. 4. Evidence for a reversal. Solid (open) symbols denote positive (negative) inclinations. Dots (or circles) are n.r.m. directions. The great circle passes through the cleaned site mean direction (solid triangle), its reversed direction (open triangle), and the mean (cross) of the reversed n.r.m. specimens. The star is the position of the earth's present field. The a.f. demagnetization curves of specimens 3 and 5 are shown in Fig. 2.

There are no results from the immediate sampling areas, but a Rb–Sr result from the Penrhyn Group, about 130 km northeast of the closest sites (1 and 2), yields an age of 2100 m.y. (Jackson and Taylor 1972). The age limits as defined by the local evidence are therefore 1250 and 2100 m.y. Closer limits are suggested later when the results are compared with those from elsewhere in the Canadian Shield.

Errors Due to Anisotropy

The Group *A* directions have been shown to be stable. Since the rocks generally are foliated the question arises as to whether this anisotropy has caused the direction of magnetization to be deflected away from the field in which it was acquired. The foliation in the sampling areas as determined from the geological maps generally dips steeply to the south (dips vary from 15 to 80° and strikes from 0 to 170°). The mean foliation is 068°, 51° SE (16 values) and this plane is at a minimum of only 10° from the mean Group *A* direction. The dispersion of the poles of the planes of foliations is however large ($k = 2.5$), greatly in excess of the dispersion of the magnetization directions (Table 2, $k = 18$), so that there is no simple correlation between remanence directions and foliations. But hidden systematic effects may be present, and in order to detect them the method of Irving and Park (1973, in preparation) is used.

One specimen from each site was reoriented into its position at the outcrop and a strong field (9000 Oe) applied first in the direction of the observed remanence. Specimens were then measured and progressively demagnetized, and directions were combined according to standard statistical procedures. The experiment was repeated but applying a field at 60° to the first one and at a large angle to the mean foliation. The effect of anisotropy in individual specimens is measured by Fisher's estimate of precision, and the presence of systematic effects by measuring departures of the mean from the applied field (Table 4).

Individual departures are often large (30°), but are usually reduced by a.f. cleaning (average departure reduced from 10° to 8°). The mean direction of the high coercive force components is 5° from the applied field. This departure exceeds the error by 1.2° (Table 4). A small but significant error due to anisotropy is present. Therefore the error in direction (α_{95} of Table 2) is adjusted by taking the root mean square of the two errors. The effect is small and a more sophisticated analysis of the errors is not warranted geophysically. Corrections for anisotropy to the error in the mean of the site poles (entry 2, Table 2) would require that estimates of error due to anisotropy be made at each site. This is a very tedious task and since the effect is small the effort was not deemed worthwhile. Accordingly, the best summary of the data is considered to be entry 1 of Table 2 with the adjusted error in brackets.

Discussion

The poles for Groups *A* and *B* are given in Table 2 and plotted in Fig. 5. Earlier in this paper it was argued that the age of the Group *A* magnetization was between 1250 and 2100 m.y. In Fig. 5 the Group *A* pole lies between the pole of the well-dated Dubawnt Group (1725 m.y.; Park *et al.* 1973) and the average of the tight cluster of poles at about 1425 m.y.

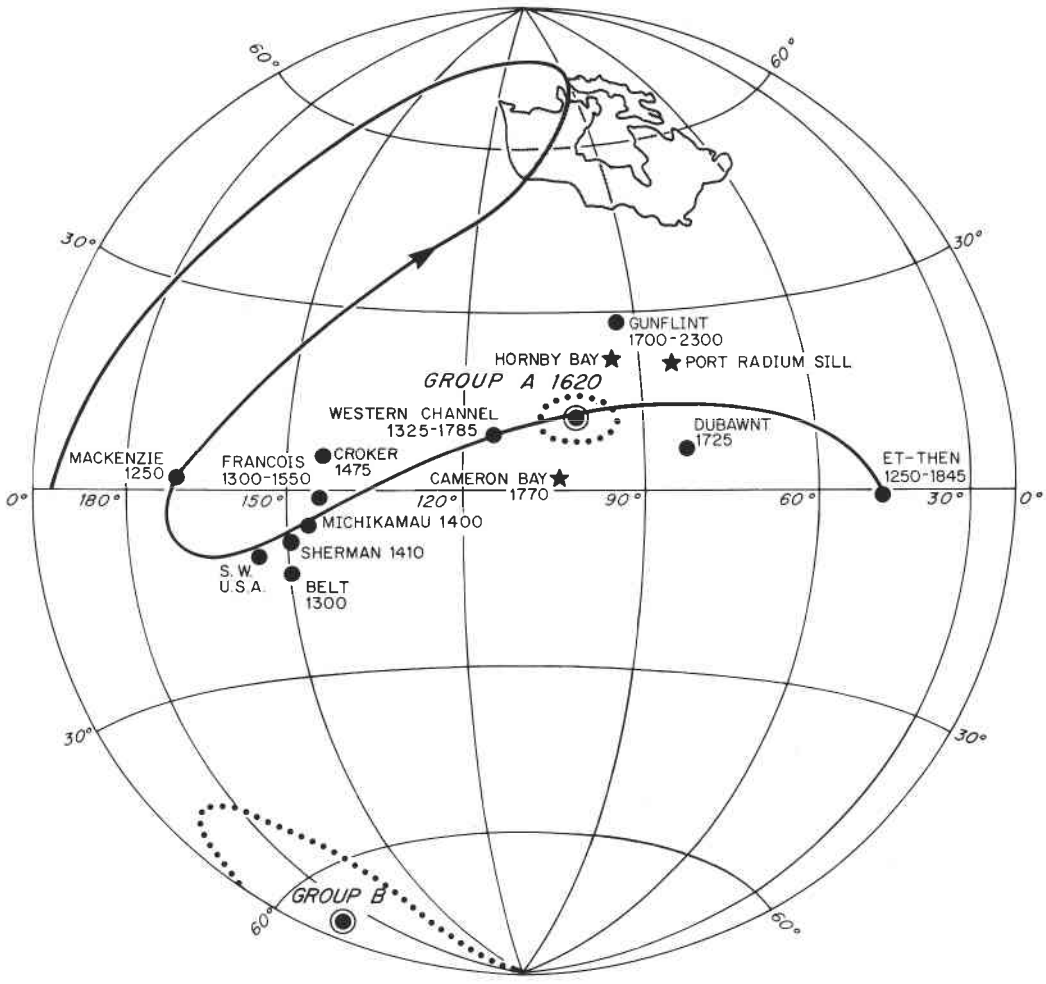


FIG. 5. Tentative Precambrian polar wander curve for the Canadian Shield (shaded) after Irving *et al.* (1972*a*). Solid dots are the results of many sites and open symbols are the results from single sites (reported in Irving *et al.* (1972*a*)). The Upper Proterozoic pole path as developed by Du Bois (1962), Black (1963), and Robertson and Fahrig (1971) is shown, but only results defining the Lower Proterozoic pole path (solid line) are plotted. Results of the present work are plotted as dots with exscribed circles, and the associated errors as ellipses of small dots. The poles plotted are Et-Then Group (Irving *et al.* 1972*b*); Dubawnt Group (Park *et al.* 1973); Gunflint Formation (Symons 1966); Croker Island Complex (Palmer 1969); Michikamau anorthosite (Murthy *et al.* 1968); Beltian rocks (average of results of Collinson and Runcorn (1960) and Norris and Black (1961)); sediments of southwestern U.S.A. (average of results of Collinson and Runcorn (1960)); St. Francois igneous rocks (average of results of Chi-Hsu *et al.* (1969) and Hays and Scharon (1966)); Mackenzie igneous rocks (average of results of Larochelle (1966), Fahrig and Jones (1969), Robertson (1969), and Irving *et al.* (1972*b*)). Arrow points in direction of polar wander.

This narrows the age limits further. Assuming a uniform rate of apparent polar wander along the earlier curve of Irving *et al.* (1972*a*) the pole of the Group *A* metamorphic rocks would be expected to have an age of about 1640 m.y. This is not significantly different from the mean K–Ar age (1622 ± 30 m.y.) and suggests that the Group *A* magnetization may be of this age. One objection to this argument would be that the sampling localities from which these paleomagnetic poles were obtained (Southern Province, Colorado, Labrador, Bear Province, etc.)

TABLE 4. Summary of anisotropy statistics for Group *A* sites

Description	Treatment (Oe)	*N*	Applied field (A)	Observed field (0)			$\lvert A - O \rvert$
			$D°, I°$	$D°, I°$	k	α_{95}	δ
Site MP19	i.r.m.	10	193, +53	193.2, +54.3	142	4.1	1.3
	500	10		194.2, +53.5	428	2.3	0.9
Site MP19	i.r.m.	10	193, −07	185.5, −03.5	244	3.1	8.3
	500	10		194.7, −04.1	195	3.5	3.3
All sites	n.r.m.	16		205.6, +74.1	5	18.3	19.6
	cleaned	16	196, +55*	190.7, +52.4	30	6.9	4.0
All sites	i.r.m.	16	196, +55	192.0, +57.3	39	6.0	3.1
	cleaned	16		188.6, +57.9	97	3.8	5.0

NOTES: *N* is the number of specimens used. One specimen was used from each site in the last four entries. δ is the angular departure of the mean observed field from the applied field. Other symbols are explained in previous tables.
*Group *A* average direction.

have moved relative to one another. Both the paleomagnetic evidence of coherent results and the geological evidence of the stabilization of this region after the earlier Hudsonian Orogeny suggest the opposite. However, much more paleomagnetic evidence is required before this can be argued with confidence.

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